7 Photons and phonons

In this section we will derive the partition function for quantised waves; the waves could be light (photons) or sound waves in a solid (phonons), approximately "spin waves" in a ferromagnet, etc. The amplitude of each allowed mode of the wave is quantised with an integer, say n, where n corresponds to "number of photons", or "number of phonons" etc as appropriate.

The ideas will be used when we come to quantum gases as well.

A common feature with all the waves is that the number of quanta (photons etc) is not fixed. The *average* number of quanta is determined by the temperature.

Firstly we need to be able to label and count the allowed waves in the system (intuitively like labelling the modes of vibration on a string or sound waves in an organ pipe).

7.1 Labelling and counting states for waves

We will assume that we may use periodic boundary conditions. (See appendix for a brief discussion.) This process is the same for all types of waves, with one qualification stated at the end of this subsection.

Initially consider one dimension. The amplitude of the wave is $u(x,t)$, at point x at time t. Periodic boundary conditions are $u(x+L,t) = u(x,t)$ for a period L. This implies, using the complex solutions to the wave equation,

$$\begin{aligned} & & u(x,t) &= \mathrm{e}^{\mathrm{i}(kx-\omega t)} \\ u(x+L,t) = u(x,t) \quad & \Rightarrow & \mathrm{e}^{\mathrm{i}(k[x+L]-\omega t)} &= \mathrm{e}^{\mathrm{i}(kx-\omega t)} \\ & \Rightarrow & \mathrm{e}^{\mathrm{i}(kL} &= 1 = \mathrm{e}^{\mathrm{i}n2\pi} \\ \text{(Here } n \text{ is any integer.)} \quad & \Rightarrow & kL &= 2\pi n \\ & \Rightarrow & k &= \frac{n2\pi}{L} \ . \end{aligned}$$

The spacing of the allowed modes is

$$\Delta k = \frac{2\pi}{L} \ ,$$

and there are two modes in that interval ($\pm 2\pi n/L$).

We may repeat this analysis in two or three dimensions with

$$u(x+L,y,z) = u(x,y,z) \quad \text{and} \quad u(x+L,y+L,z) = u(x,y,z) \ , \ \text{etc., etc.}$$

From this, upon substitution of

$$u(x,y,z) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} = \mathrm{e}^{\mathrm{i}(k_x x + k_y y + k_z z)} \ ,$$

we see that

$$\mathbf{k} = (k_x, k_y, k_z) = \frac{2\pi}{L}(m_x, m_y, m_z) = \frac{2\pi}{L}\mathbf{m} \ ,$$

where m_x, m_y and m_z are any three integers (positive, negative or zero).[1]

As mentioned at the start of this subsection, this process is the same for most descriptions of waves. The provisos are that: (i) one is describing travelling not standing waves (in the latter case only positive integers are allowed); (ii) waves may travel in both directions – if not, then again a restriction to positive integers occurs.

7.2 The continuum limit

We already observed that to calculate the partition function for a quantum particle in a box is difficult. If the box becomes very large we will argue we may replace the awkward sums in the partition function by integrals which allow Z to be calculated.

This part of the description of waves **is sensitive** to the form of the dispersion relation, $\omega(\mathbf{k})$, in the change of variables from $\mathbf{k}$ to ω. An example where $\omega(\mathbf{k}) \propto k^2$ is on problem sheet 3.

See Fig. (7.1), for a depiction of the allowed modes in two dimensions. The important point is the area per mode in "k-space" is $(2\pi/L)^2$ in two dimensions and $(2\pi/L)^3$ in three dimensions.

The consequence of the observation about the area per mode is that (say for three dimensions) we may replace counting the modes in a region of k-space by an integral over k-space divided by the volume per mode I.e.

$$\sum_{\mathbf{m}} 1 \simeq \frac{\int \mathrm{d}\mathbf{k}}{(2\pi/L)^3} = \left(\frac{L}{2\pi}\right)^3 \int \mathrm{d}k_x \mathrm{d}k_y \mathrm{d}k_z = n_k \int \mathrm{d}k_x \mathrm{d}k_y \mathrm{d}k_z \ . \tag{7.1}$$

The interpretation of the factor $n_k = (L/(2\pi))^3$ is of the density of states in k-space, equivalently the reciprocal of the (k-space) volume per mode. In general,

$$n_k^{(d)} = \left(\frac{L}{2\pi}\right)^d$$

[1] Note that $m_x = m_y = m_z = 0$ is a special case, with $\omega = 0$. We ignore this as it gets lost when we take limit $L \to \infty$.

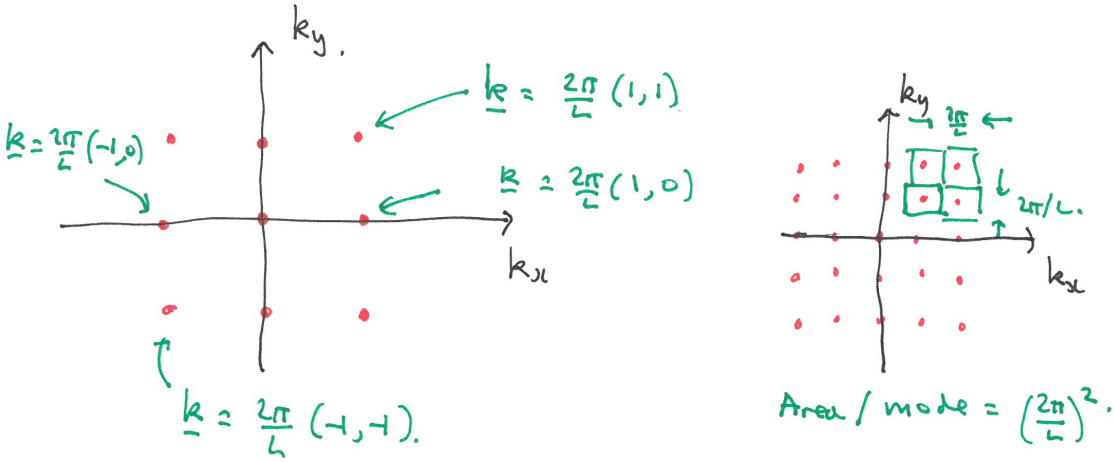

Figure 7.1: Two dimensional case. Left figure: the allowed values of $\mathbf{k}$ from periodic boundary conditions. Right figure: area per mode (green squares) is $(2\pi/L)^2$.

is the density of modes in d-dimensions with $n_k^{(3)} = n_k$ defined above.

Of course Eq. (7.1) is **not exact** but the discrepancy between counting and integration becomes negligible as $L^3 \to \infty$, i.e. for large systems, where the volume per mode tends to zero. (But see the Appendix for some qualification.)

7.3 One mode as a harmonic oscillator

Let us consider one dimension for simplicity. The wave amplitude, $u(x,t)$, obeys the wave equation (in the simplest case):

$$\frac{1}{\mathrm{c}^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} \qquad \text{Let: } u(x,t) = \mathrm{e}^{\mathrm{i}kx} U_k(t) \ , \ \text{(I.e. one mode, } k.\text{)}$$

Upon substitution into the wave equation we see that

$$\begin{aligned} \frac{1}{\mathrm{c}^2}\ddot{U}_k\,\mathrm{e}^{\mathrm{i}kx} &= -k^2 U_k(t)\,\mathrm{e}^{\mathrm{i}kx} \\ \Rightarrow \quad \ddot{U}_k &= -\mathrm{c}^2 k^2\, U_k \ , \end{aligned}$$

which should be compared with $\ddot{x} = -\omega_0^2 x$, with $\omega_0 = \sqrt{K/m}$.

Thus we see a simple harmonic oscillator emerges for the quantity $U_k(t)$, describing one mode with natural frequency

$$\omega(k) = \mathrm{c}k \ .$$

Thus we may quantise the waves using the process from the last section, one mode at a time.

7.4 Energy of quantised waves

Consider one mode, labelled by $\mathbf{m}$:

$$\mathbf{k} = \frac{2\pi}{L}\mathbf{m} \ ,$$

with frequency $\omega_{\mathbf{m}}$. We regard the mode as a harmonic oscillator, as derived in the last subsection.

Then the energy in the mode is

$$E_{\mathbf{m}}(n_{\mathbf{m}}) = \left(n_{\mathbf{m}} + \tfrac{1}{2}\right)\hbar\omega_{\mathbf{m}} \ ,$$

where $n_{\mathbf{m}}$ is the number of photons in the mode. The total energy, $E^{\text{tot}}(\{n_{\mathbf{m}}\})$, depending on the numbers of photons in all modes, follows:

$$\begin{aligned} E^{\text{tot}}(\{n_{\mathbf{m}}\}) &= \sum_{\mathbf{m}} E_{\mathbf{m}}(n_{\mathbf{m}}) \\ &= \sum_{\mathbf{m}} \left(n_{\mathbf{m}} + \tfrac{1}{2}\right)\hbar\omega_{\mathbf{m}} \ . \end{aligned}$$

Remembering the expression for the number of states, $W_{\text{sing}}(E)$, of energy E for the single particle quantum system with energy levels E_n with degeneracy g_n:

(Sum over levels, n)
$$W_{\text{sing}}(E) = \sum_n g_n \delta\left(E - E_n\right)$$

(Sum over *states*, i)
$$= \sum_i \delta\left(E - E_i\right) \ ,$$

Where the sum over *states* we include every state explicitly, as against grouping them into degenerate *levels*. Another way of stating this is that all the energy levels, E_n are distinct, whereas the states, E_i, may have identical values of energy.

Then we may write the number of states of the quantised optical field, $W_{\text{opt}}(E)$, as

$$\begin{aligned} W_{\text{opt}}(E) &= \sum_{\{n_{\mathbf{m}}\}} \delta\left(E - \sum_{\mathbf{m}} E_{\mathbf{m}}(n_{\mathbf{m}})\right) \\ &= \sum_{\{n_{\mathbf{m}}\}} \delta\left(E - \sum_{\mathbf{m}} \left(n_{\mathbf{m}} + \tfrac{1}{2}\right)\hbar\omega_{\mathbf{m}}\right) \ , \end{aligned}$$

where the sum **runs over all choices** for the number of photons in each mode, $n_{\mathbf{m}}$.

In **one** dimension, where $\mathbf{m}$ is replaced by the single integer m, the explicit form of the sum would be:

$$\sum_{\{n_m\}} = \cdots \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots \sum_{n_m=0}^{\infty} \cdots \ .$$

The three-dimensional case is less easily written down explicitly!

7.5 Exponential of a sum and a product of exponentials

In this section we generalise the result

$$\mathrm{e}^{a+b} = \mathrm{e}^{a}\mathrm{e}^{b} \ .$$

Consider the following sum:

$$\sum_{m=0}^{\infty} a_m = a_0 + a_1 + a_2 + \cdots + a_m + \cdots$$

Then we may write

$$\begin{aligned} \exp\left\{\sum_{m=0}^{\infty} a_m\right\} &= \exp\left\{a_0 + a_1 + a_2 + \cdots + a_m + \cdots\right\} \\ &= \mathrm{e}^{a_0}\mathrm{e}^{a_1}\mathrm{e}^{a_2}\cdots\mathrm{e}^{a_m}\cdots \\ \Rightarrow \quad \exp\left\{\sum_{m=0}^{\infty} a_m\right\} &= \prod_{m=0}^{\infty} \exp\left\{a_m\right\} \ . \end{aligned} \tag{7.2}$$

Although easier to see explicitly with lower limit of $m = 0$ easy to generalise to lower limit being $m = -\infty$. There:

$$\exp\left\{\sum_{m=-\infty}^{\infty} a_m\right\} = \prod_{m=-\infty}^{\infty} \exp\left\{a_m\right\} \ .$$

And we may generalise to the three-dimensional sum over $\mathbf{m} = (m_x, m_y, m_z)$:

$$\begin{aligned} \exp\left\{\sum_{\mathbf{m}} a_{\mathbf{m}}\right\} &= \prod_{\mathbf{m}} \exp\left\{a_{\mathbf{m}}\right\} \\ &= \prod_{m_x=-\infty}^{\infty} \prod_{m_y=-\infty}^{\infty} \prod_{m_z=-\infty}^{\infty} \exp\left\{a_{(m_x,m_y,m_z)}\right\} \ . \end{aligned} \tag{7.3}$$

7.6 Photon partition function

We may now use the usual expression for the canonical partition function:

$$\begin{aligned} Z_{\mathsf{opt}}(T,V) &= \int \mathrm{d}E \ W_{\mathsf{opt}}(E)\,\mathrm{e}^{-\beta E} \\ &= \int \mathrm{d}E \ \sum_{\{n_{\mathbf{m}}\}} \delta\left(E - \sum_{\mathbf{m}} \left(n_{\mathbf{m}} + \tfrac{1}{2}\right)\hbar\omega_{\mathbf{m}}\right) \mathrm{e}^{-\beta E} \\ \text{(Allowing delta function to act.)} \quad &= \sum_{\{n_{\mathbf{m}}\}} \exp\left\{-\beta \sum_{\mathbf{m}} \left(n_{\mathbf{m}} + \tfrac{1}{2}\right)\hbar\omega_{\mathbf{m}}\right\} \end{aligned}$$

We may now disentangle the distinct modes in the argument of the exponential, using Eq. (7.3):

$$\begin{aligned} Z_{\text{opt}}(T,V) &= \sum_{\{n_{\mathbf{m}}\}} \exp\left\{-\beta \sum_{\mathbf{m}} \left(n_{\mathbf{m}} + \tfrac{1}{2}\right) \hbar\omega_{\mathbf{m}}\right\} \\ &= \sum_{\{n_{\mathbf{m}}\}} \prod_{\mathbf{m}} \exp\left\{-\beta \left(n_{\mathbf{m}} + \tfrac{1}{2}\right) \hbar\omega_{\mathbf{m}}\right\} \\ &= \prod_{\mathbf{m}} \left[\sum_{n_{\mathbf{m}}=0}^{\infty} \exp\left\{-\beta \left(n_{\mathbf{m}} + \tfrac{1}{2}\right) \hbar\omega_{\mathbf{m}}\right\}\right] \end{aligned}$$

In the last line we decompose the sum over the numbers of photons in all modes into the sums appropriate to each member of the product over modes. We may now perform the geometric series as we did for the single harmonic oscillator.

$$Z_{\text{opt}}(T,V) = \prod_{\mathbf{m}} \frac{\mathrm{e}^{-\beta\hbar\omega_{\mathbf{m}}/2}}{1-\mathrm{e}^{-\beta\hbar\omega_{\mathbf{m}}}} \ ,$$

which allows us to write down the Helmholtz free energy:

$$\begin{aligned} F_{\text{opt}}(T,V) &= -k_{\text{B}}T \ln Z_{\text{opt}}(T,V) \\ &= -k_{\text{B}}T \sum_{\mathbf{m}} \left\{-\tfrac{1}{2}\beta\hbar\omega_{\mathbf{m}} - \ln\left(1-\mathrm{e}^{-\beta\hbar\omega_{\mathbf{m}}}\right)\right\} \\ &= \sum_{\mathbf{m}} \left\{ \underbrace{\tfrac{1}{2}\hbar\omega_{\mathbf{m}}}_{\text{zero point energy}} + k_{\text{B}}T \ln\left(1-\mathrm{e}^{-\beta\hbar\omega_{\mathbf{m}}}\right) \right\} \end{aligned}$$

Note that the zero point energy provides an infinite constant – due to the sum over all modes, $\mathbf{m}$. Surprisingly, perhaps, this does not make it unphysical: for example one can measure *changes* in the zero-point energy as the geometry changes (this leads to the Casimir effect, a generalisation of the van der Waals interaction). More speculatively, whether the zero-point energy has a gravitational effect is not understood. We will discard the zero point term from now on. The term does not depend on T, so the partial derivatives yielding entropy, S and temperature dependence of pressure, p, are not affected by the term. It serves to define an (infinite) zero of internal energy, U, F and the other thermodynamic potentials.

We may now replace the sum over $\mathbf{m}$, by an integral as in Eq. (7.1) (which becomes increasingly accurate as $L \to \infty$) to obtain (a factor of 2 for the two transverse polarisations of electromagnetic waves has been included as well as the factor, n_k, for the density of modes in

k-space)[2]

$$\begin{aligned} F_{\text{opt}}(T,V) &= 2n_k^{(3)}\, k_{\text{B}}T \int \mathrm{d}k_x \int \mathrm{d}k_y \int \mathrm{d}k_z \ln\left(1-\mathrm{e}^{-\beta\hbar\omega_{\mathbf{k}}}\right) \\ &= 2\left(\frac{L}{2\pi}\right)^3 k_{\text{B}}T \int \mathrm{d}k_x \int \mathrm{d}k_y \int \mathrm{d}k_z \ln\left(1-\mathrm{e}^{-\beta\hbar\omega_{\mathbf{k}}}\right) \\ \text{(Change to spherical polars)} \quad &= 2\left(\frac{L}{2\pi}\right)^3 k_{\text{B}}T \underbrace{4\pi}_{\text{from angular integrals}} \int k^2 \mathrm{d}k \ln\left(1-\mathrm{e}^{-\beta\hbar\omega_{\mathbf{k}}}\right) \ . \end{aligned} \tag{7.4}$$

We have defined spherical polar coordinates, (k, θ_k, ϕ_k), in $\mathbf{k}$-space in a similar manner as in real space:

$$k_x = k\sin\theta_k\cos\phi_k \ ; \ k_y = k\sin\theta_k\sin\phi_k \ ; \ k_z = k\cos\theta_k \ .$$

The 4π in Eq. (7.4) originates from the integrals over θ_k and ϕ_k, to give the area of the sphere with radius $k = 1$.

Now let us change from an integral over wave-vector, k, to an integral over

$$x = \beta\hbar\omega \quad \Rightarrow \quad \omega = \frac{k_{\text{B}}Tx}{\hbar} \quad \text{and} \quad k = \frac{\omega}{\mathrm{c}} = \frac{k_{\text{B}}Tx}{\hbar\mathrm{c}} \ ,$$

where c is the speed of light.

We find (noting $(2\pi)^3\hbar^3 = h^3$):

$$F_{\text{opt}}(T,V) = 8\pi\frac{(k_{\text{B}}T)^4}{h^3\mathrm{c}^3}V \int_0^\infty \mathrm{d}x\, x^2 \ln\left(1-\mathrm{e}^{-x}\right)$$

The integral over x is *negative* as the argument of the log is less than one for all x. It may be performed by one integration by parts and then Taylor expanding the denominator and using a Fourier series to sum the result (see Baierlein p422, table A2). Then:

$$F_{\text{opt}}(T,V) = -\frac{8\pi^5}{45}\frac{(k_{\text{B}}T)^4}{h^3\mathrm{c}^3}V \ . \tag{7.5}$$

We may now deduce the entropy, internal energy and pressure as usual from F. The pressure is especially easy:

$$p = -\left(\frac{\partial F}{\partial V}\right)_T = \frac{8\pi^5}{45}\frac{(k_{\text{B}}T)^4}{h^3\mathrm{c}^3} = -\frac{F}{V} \ .$$

This expression for the "radiation pressure" is useful in understanding dust in the Solar System (see problem sheet 3), and in the *Eddington limit* of the luminosity of a body – where the gravitational attraction of a star, accreting black hole etc balances the radiation pressure out

[2] The accuracy of the replacement of the sum by an integral can be quantified. The two-dimensional result (Gauss' Circle problem) is a classical result where the the difference between the sum and integral over a circular region of radius r scales as r^α, where $\frac{1}{2} < \alpha \leq 131/208$. In higher dimensions there are also some results.

of the star. (But this can be exceeded for a period of time e.g. "A super-Eddington-accreting black hole". Suh et al *Nature Astronomy*, , (2024).)

There is a curious feature about the pressure: there is no dependence on the volume. This seems peculiar by comparison with an ideal gas. Firstly because the equation of state is just the function $p(T)$ and not $p(T, V)$. Secondly if one considers an isothermal process for an ideal gas, $p = nRT/V$, so the pressure drops if V increases.

Why is there not a drop in pressure as one increases, isothermally, the volume which contains the photons?

The average *number* of photons in *each mode* is independent of volume. A simple picture is of the photons bouncing between opposite walls, with the reflections causing pressure (due to the impulse from the reversal of the component of momentum, $\hbar\mathbf{k}$, perpendicular to the wall, upon reflection). Then if one doubles the distance between the walls, $L \to 2L$, the number of reflections per unit time will drop by a factor of two – due to the time between collisions being $2L/\mathrm{c}$. So surely the pressure drops?

But, the number of allowed modes is *not* constant. Let us repeat the periodic boundary condition calculation of the modes: we find:

$$k = \frac{n2\pi}{L} \to k = \frac{n2\pi}{2L} \ ,$$

so there is a new mode between each of those for L. Thus the number of photons in equilibrium doubles – due to the number of modes being populated thermally doubling. This exactly compensates for the decrease of frequency of reflections.

Thus the pressure remains constant, and hence the isothermal compressibility of the photon gas,

$$\kappa_{\mathrm{T}} = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T \ ,$$

is infinite. This has been demonstrated experimentally – albeit with (interesting) complications – in 2022, Busley, Erik *et al*, *Science*, **375**, 1403 (2022). There is an associated less technical article here.

For the case of an *adiabatic* expansion, see problem sheet 3, where the Cosmic Microwave Background – which is blackbody radiation at $T \simeq 2.7\,\mathrm{K}$, see Fig. (7.2), is an example.

The results on the photon gas may be related to the Stefan-Boltzmann law. The power, P, radiated by an area A is

$$P = \frac{1}{4}\frac{U}{V}A\,\mathrm{c} = \sigma A T^4 \quad \text{where } \sigma = \frac{2\pi^5 k_{\mathrm{B}}^4}{15\mathrm{h}^3\mathrm{c}^2} \ ,$$

and σ is the Stefan-Boltzmann constant. The factor of 1/4 comes from kinetic theory: the flux depends of the angle of approach by the photons to the area, ie velocity normal to the

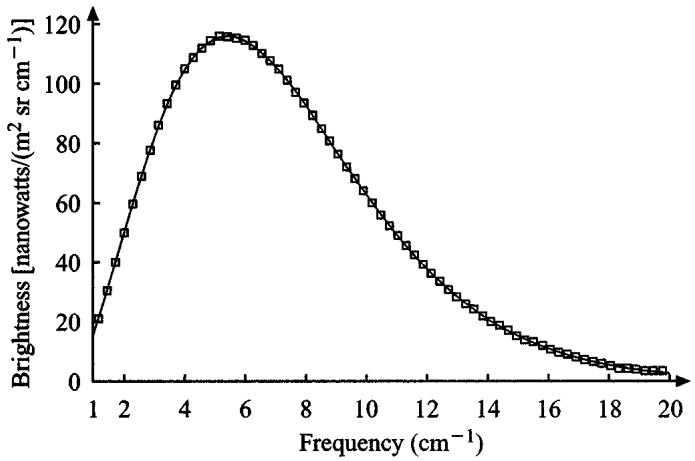

Figure 7.2: The spectrum of the Cosmic Microwave Background (CMB), the horizontal, "frequency", axis is in units of ν/c, where ν is the frequency. Credit from J.C. Mather et al, *Astrophys. J.* **354**, L37 (1990) via Baierlein *Thermal Physics*, chapter 6.

area is $\mathrm{c}\cos\theta$, where θ is angle to the normal. Then to average this over the directions - assuming the distribution of photon wave vectors is uniform, the net flux is

$$\begin{aligned}\int_0^{\pi/2} \cos\theta \frac{\mathrm{d}\Omega}{4\pi} &= \frac{1}{4\pi}\int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi/2} \cos\theta \sin\theta \mathrm{d}\theta \\ &= \frac{1}{2}\int_0^{\pi/2} \cos\theta \sin\theta \mathrm{d}\theta \\ &= \frac{1}{4}\ .\end{aligned}$$

7.7 Phonons

Phonons are quanta of sound waves in a solid (or a quantum liquid – e.g. helium for $T < 2\mathrm{K}$ – or a quantum gas such as sodium or rubidium).

To a first approximation the calculation of the free energy parallels that of the photon gas – replacing the speed of light by the speed of sound.

But there are some differences:

1. The dispersion relation, $\omega(k)$, while being linear, $\omega \simeq \mathrm{c}_{\mathsf{sound}} k$, for "small" wave vectors, $k \ll \pi/a$ where a is the spacing of atoms in the solid, curves over for $k \sim \pi/a$.
2. There is a maximum wave vector which depends on the direction of the wave vector.
3. In three dimensions, for a solid, there are three branches of the dispersion relation, with

distinct speeds of sound for the *two* transverse modes, $\mathrm{c_t}$ and the *single* longitudinal mode, $\mathrm{c_l}$ with $\mathrm{c_l} > \mathrm{c_t}$.

4. As well as the *acoustic* branches mentioned in point 3, there are also *optical branches*, at higher frequencies – in general 3 branches for every atom in the primitive unit cell.

5. Interactions between the phonons cause some effects, for example: thermal expansion, finite thermal conductivity in an insulator.

We simplify to the following *Debye* model.

1. The dispersion relation for the three polarisations of phonons is assumed to be identical: $\omega(k) = \bar{\mathrm{c}}k$, where the average speed of sound, $\bar{\mathrm{c}}$, is defined in problem sheet 3.

2. There is a limit to the maximum k (or shortest wavelength), denoted $k_{\mathrm{D}} \geq k$, the Debye wave vector.

3. We assume the dispersion relation is linear for $0 \leq k \leq k_{\mathrm{D}}$, $\omega(k) = \bar{\mathrm{c}}k$.

4. We assume there are no interactions between the phonons. (At low temperatures, this is a good approximation as the density of phonons – regarded as a gas – is very low, so probability of collision is small).

Here, we will focus on the low temperature limit. We define low temperature to be $\hbar\bar{\mathrm{c}}k_{\mathrm{D}} \gg k_{\mathrm{B}}T$, so the phonons with wave vectors near the edge of the allowed region, $k \simeq k_{\mathrm{D}}$, are almost never thermally excited. We may then replace the upper limit in the integral over k by ∞ instead of k_{D}. We then obtain F by direct analogy with Eq. (7.5). We replace c, the speed of light by the "average" speed of sound, $\bar{\mathrm{c}}$ (actually a little more complicated – see problem sheet 3), and replace a factor of 2 for the two electromagnetic polarisations for 3 for the three phonon polarisations (if we are dealing with a solid or 1 we were dealing with a quantum liquid/gas):

$$\begin{aligned} F_{\text{phonon}}(T,V) &= \frac{3}{2}\frac{\mathrm{c}^3}{\bar{\mathrm{c}}^3}F_{\text{opt}}(T,V) \\ &= -\frac{3}{2}\frac{8\pi^5}{45}\frac{(k_{\mathrm{B}}T)^4}{h^3\bar{\mathrm{c}}^3}V \\ &= -\frac{4\pi^5}{15}\frac{(k_{\mathrm{B}}T)^4}{h^3\bar{\mathrm{c}}^3}V \ . \end{aligned}$$

Then we easily obtain:

$$\begin{aligned} U_{\text{phonon}} &= \frac{\partial(\beta F_{\text{phon}})}{\partial\beta} \\ &= -\frac{4\pi^5}{15}\frac{\partial}{\partial\beta}\frac{\beta^{-3}}{h^3\bar{\mathrm{c}}^3}V \\ &= (-1)(-3)\frac{4\pi^5}{15}\frac{\partial}{\partial\beta}\frac{\beta^{-4}}{h^3\bar{\mathrm{c}}^3}V \\ &= \frac{4\pi^5}{5}\frac{(k_{\mathrm{B}}T)^4}{h^3\bar{\mathrm{c}}^3}V \ . \end{aligned}$$

And then the famous *Debye specific heat*:

$$C_V^{\text{phonon}} = \left(\frac{\partial U}{\partial T}\right)_V$$
$$= \frac{16\pi^5}{5}\,\frac{k_{\text{B}}^4 T^3}{h^3 \bar{c}^3} V \ .$$

Previously Einstein had proposed a model where all the atoms oscillated at the same frequency – but that yielded an exponentially small specific heat at low temperatures. The earlier Dulong-Petit theory paralleled a gas – i.e. a constant specific heat. An example of how well this works experimentally is in Fig. (7.3).

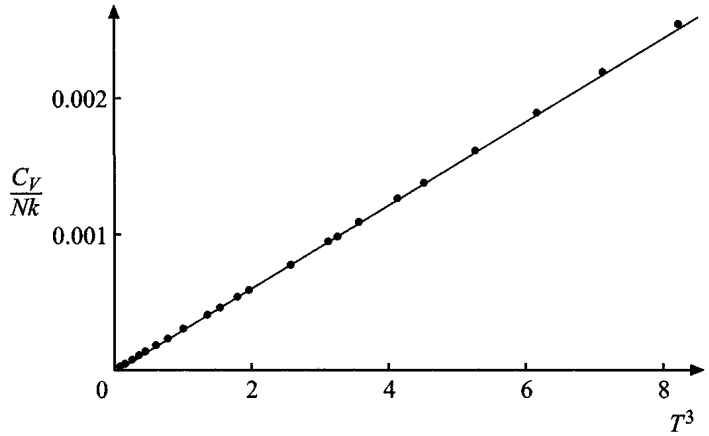

Figure 7.3: The specific heat of solid Argon for $0 \leq T \leq 2\,\text{K}$. **Note** the horizontal axis is T^3, which is why the graph is a straight line.

Understanding the contribution which phonons make to C_V, one may extract any other anomalous behaviour by subtracting the appropriate Debye expression which only requires a measurement of the speeds of sound. An example is in Fig. (7.4), from J. Goraus et al, *J. Phys. Cond. Mat.* **25**, 176002 (2003). The upturn in the specific heat indicates that this material contains more ingredients than merely phonons – indeed there are both aspects of magnetic order and superconductivity at low temperatures.

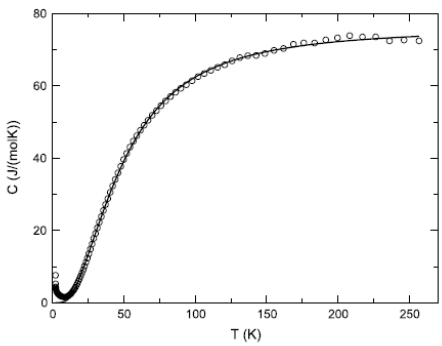

Figure 7.4: The specific heat, C, for the compound CdPdBi, showing Debye behaviour down to $10\,\text{K}$, and anomalous behaviour for $T < 10\,\text{K}$.

The specific heat has a unique importance as an unbiased measure of order occurring in a system. This is in contrast to, for example, magnetic susceptibility indicating potential magnetic order and resistivity indicating superconductivity. The most famous example is the

"hidden order" phase first identified in URu_2Si_2 in Palstra, T.T.M et al *Phys. Rev. Lett.*, **55**, 2727 (1985), see Fig. (7.5), which has still not been clarified after 36 years of study. A recent review on this very interesting material is J.A. Mydosh et al *J. Phys.: Condens. Matter* **32** 143002 (2020).

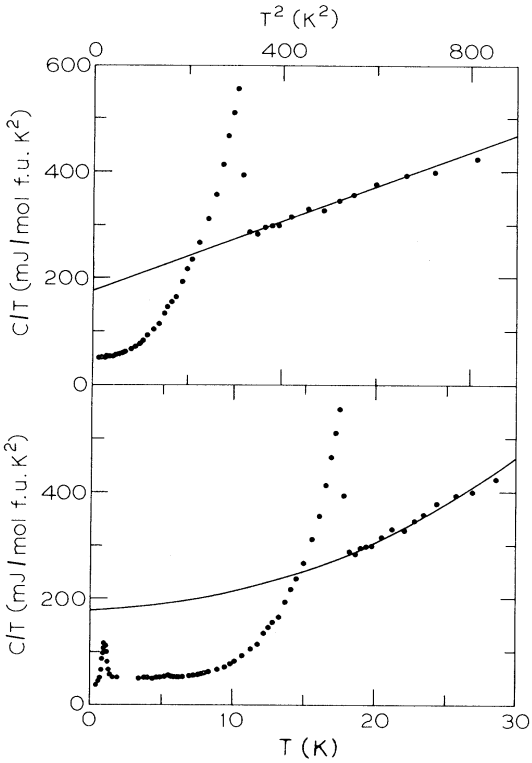

Figure 7.5: The specific heat of URu_2Si_2, plotted in a manner to extract the phonon, Debye, terms and ordinary electronic terms from the measured specific heat. The peak indicates the still-mysterious phase transition. Note the difference of horizontal variables. Credit: Palstra et al.

One trick about Fig. (7.5) is the choice of both the "x-" and "y-axes" in the figure. As we will see in a few lectures' time, the specific heat of a fermi gas (eg electrons in a conductor) at low temperatures is proportional to T. The phonons provide a T^3 term. To separate these – and reveal anything anomalous – it is convenient to write:

$$C = AT + BT^3$$

$$\Rightarrow \quad \frac{C}{T} = A + BT^2 \tag{7.6}$$

(Let $x = T^2$ and $y = C/T$.) $$\Rightarrow \quad y = A + Bx \ . \tag{7.7}$$

The lower part of Fig. (7.5) corresponds to Eq. (7.6) whereas the upper part of Fig. (7.5) corresponds to Eq. (7.7). So one can read off, in upper part of Fig. (7.5), A as being the intercept and B as being the gradient of Fig. (7.5).

Finally one may interpret the form of the upper part of Fig. (7.5) in a little more detail. Remember that, at constant volume:

$$đQ = C_V \mathrm{d}T = T\mathrm{d}S$$

$$\Rightarrow \quad \int \frac{C_V}{T}\mathrm{d}T = \int \mathrm{d}S \ .$$

Thus the peak in the C/T-T^2 graph above the straight line (which corresponds to the standard electron and phonon contributions), is entropy being removed at a **higher** temperature than in a "normal" system, so one may say the system is "ordering", and the singularity at the higher temperature end of the peak represents a phase transition. There is a corresponding decrease in the entropy at lower temperatures – which is why the curve is depressed below the straight line there.

Appendices

A Periodic boundary conditions – non-examinable

The insensitivity to the choice of boundary conditions for solutions of, say, the 1d wave equation, $u(x,t)$ may be explored by comparing "box" boundary conditions, $u(0,t)=0$ and $u(L,t)=0$, with periodic boundary conditions $u(x+L,t)=u(x,t)$.

For periodic boundary conditions we use the complex solutions to the wave equation,

$$\begin{aligned} u(x,t) &= \mathrm{e}^{\mathrm{i}(kx-\omega t)} \\ \Rightarrow \quad \mathrm{e}^{\mathrm{i}(k[x+L]-\omega t)} &= \mathrm{e}^{\mathrm{i}(kx-\omega t)} \\ \Rightarrow \quad \mathrm{e}^{\mathrm{i}(kL} &= 1 = \mathrm{e}^{\mathrm{i}n2\pi} \\ \Rightarrow \quad kL &= 2\pi n \\ \Rightarrow \quad k &= \frac{n2\pi}{L} \ . \end{aligned}$$

The spacing of the allowed modes is

$$\Delta k = \frac{2\pi}{L} \ ,$$

and there are two modes in that interval $(\pm 2\pi n/L)$.

For the box boundary conditions we have standing waves

$$\begin{aligned} \text{(Automatic satisfaction at } x=0.) \quad u(x,t) &= \sin kx \cos \omega t \\ \text{(At } x=L.) \quad u(L,t) = 0 &= \sin kL \cos \omega t \\ \Rightarrow \quad kL &= n\pi \\ \Rightarrow \quad k &= \frac{n\pi}{L} \ , \end{aligned}$$

so the spacing is $\Delta k = \pi/L$. But there is one mode in that interval.

So the number of modes per unit wave vector is the same – with the box-allowed modes interleaving the periodic boundary condition modes.

It is thus *plausible* that changing the boundary conditions only moves the modes around on a very fine scale and the average number of modes remains the same.

Proving that statistical mechanics is insensitive to the boundary conditions in two or more dimensions is not straightforward. For example the pressure of a classical or bose gas being

independent of the boundary conditions for an arbitrary shape of container (but without cusps on the boundary) is shown in "On the independence of the thermodynamic limit on the boundary conditions in quantum statistical mechanics", *Comm. Math. Phys.*, **29**, 15 (1973).

However sometimes if you look not just at the dominant term as $V \to \infty$, one can discern differences with the shape or boundary conditions. The first results on this were by Mark Kac "Can you hear the shape of this drum?", *Amer. Math. Month* **73** p 1 (1966) . However more recently different shaped drums have been constructed with the same vibrational frequencies, they are termed isospectral. (Use the second link on Library website – to ProQuest Central – it is easier to use.) See Fig. (A.1) for a picture of Kac and here is the pronunciation of his surname.

Figure A.1: Mark Kac. Credit: Wiki